National University of Computer and Emerging Sciences

BS CS | **MT1003 Calculus and Analytical Geometry**

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Quiz : 4.6

Q1: In the pursuit of engineering a 1000 cm3 voluminous right circular cylindrical receptacle, a meticulous consideration of the waste factor during the production process is paramount. It's imperative to note that when fashioning the can's lateral surface, not a single iota of aluminum is squandered. However, an intriguing departure from convention emerges when dealing with the top and bottom components, each having a radius 'r.' These components are crafted through the meticulous art of cutting squares, with sides measuring precisely 2r units.

As a consequence of this innovative approach, the equation representing the total amount of aluminum resource invested in the creation of this can is remarkably expressed as **A = 8r^2 + 2πrh.** This, mind you, is a significant deviation from the erstwhile equation, A = 2πr^2 + 2πrh, as observed in a previous illustrative example. In said example, it was conclusively determined that the most economically viable can exhibits a unique ratio between the can's height 'h' and the can's radius 'r.' Given this novel premise, one is prompted to ponder: what is the revised and intriguing ratio in this discerning scenario? **(10)**

Answer:

With a volume of 1000 cm^3 and $V = \pi r^2 h$, then $h = \frac{1000}{\pi r^2}$. The amount of aluminum used per can is $A = 8r^2 + 2\pi rh = 8r^2 + \frac{2000}{r}$. Then $A'(r) = 16r - \frac{2000}{r^2} = 0 \Rightarrow \frac{8r^3 - 1000}{r^2} = 0 \Rightarrow$ the critical points are 0 and 5, but $r = 0$ results in no can. Since $A''(r) = 16 + \frac{1000}{r^3} > 0$ we have a minimum at $r = 5 \Rightarrow h = \frac{40}{\pi}$ and $h\!:\!r = 8\!:\!\pi$.

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